# Shannon's channel coding theorem: the direct part

Introduction to Information theory

support course material

neves

DRAFT

## Can we communicate reliably at a positive rate?

A channel flips each transmitted bit with probability $p = 0.11$, independently.

- **Send the bits as they are.** Rate $1$, but a block of $24$ bits arrives intact with probability $0.89^{24} \approx 0.061$.
- **Repeat every bit.** The error can be made tiny, but the rate $1/n$ goes to $0$.
- **Shannon (1948).** Neither sacrifice is needed: every rate below a number $C$, the *capacity*, can be used with an error probability that tends to $0$. Here $C \approx 0.50$ bits per channel use.

This lecture proves the *direct part*: every rate $R < C$ is achievable. The converse ($R > C$ is not) is the next lecture.

## Plan of the lecture

**The six parts**, as in the video:

1. Reliable transmission of information
2. Reliability by repetition, and its price
3. Noise is predictable in bulk
4. How many messages can the noise keep apart?
5. A random codebook and a threshold decoder
6. Why the error vanishes

**Prerequisites.** Entropy and mutual information, Chebyshev's inequality, the weak law of large numbers.

**Conventions.** $\log = \log_2$, information in bits. $h(q) = -q \log q - (1-q)\log(1-q)$ is the binary entropy. $\mathrm{Ber}(q)$, $\mathrm{Bin}(n, q)$: Bernoulli and binomial laws. $x^n = (x_1, \ldots, x_n)$.

**Running example.** $\mathrm{BSC}(p)$ with $p = 0.11$: $h(0.11) \approx 0.4999$, $C \approx 0.50$. Slack $\varepsilon = 0.05$, threshold $p + \varepsilon = 0.16$.

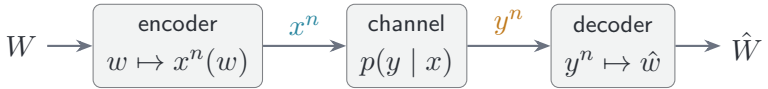
Part 1

# Reliable transmission of information

The problem of reliable transmission of information: model, rate, error, capacity, and what Shannon promised.


1 **Reliable transmission of information**
2 Reliability by repetition, and its price
3 Noise is predictable in bulk
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## The communication problem




**Definition 1.1 (Code).** An $(M, n)$ *code* for a channel with input alphabet $\mathcal{X}$ and output alphabet $\mathcal{Y}$ consists of a *message set* $\mathcal{W} = \{1, \dots, M\}$, an *encoder* $w \mapsto x^n(w) \in \mathcal{X}^n$, whose values are the *codewords* (together, the *codebook*), and a *decoder* $g : \mathcal{Y}^n \to \mathcal{W}$, which turns the received block into a guess $\hat{w} = g(y^n)$.

The decoder sees only $y^n$ and the codebook. It never sees $w$, and it never sees the noise.

## Memoryless channels

**Definition 1.2 (Discrete memoryless channel).** A *discrete channel* $(\mathcal{X}, p(y \mid x), \mathcal{Y})$ has finite alphabets and *transition probabilities* $p(y \mid x)$. Used $n$ times, it is *memoryless* if $p(y_i \mid x^i, y^{i-1}) = p(y_i \mid x_i)$, and used *without feedback* if $p(x_i \mid x^{i-1}, y^{i-1}) = p(x_i \mid x^{i-1})$, for every $i$.

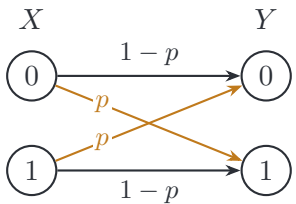
**Lemma 1.3 (Product formula).** For a memoryless channel used without feedback, $p(y^n \mid x^n) = \prod_{i=1}^{n} p(y_i \mid x_i)$ whenever $p(x^n) > 0$.

*Proof.* Chain rule, alternating inputs and outputs:

$$\begin{aligned} p(x^n, y^n) &= \textstyle\prod_{i=1}^{n} p(x_i \mid x^{i-1}, y^{i-1})\, p(y_i \mid x^i, y^{i-1}) \\ &= \textstyle\prod_{i=1}^{n} p(x_i \mid x^{i-1})\, p(y_i \mid x_i) = p(x^n) \prod_{i=1}^{n} p(y_i \mid x_i), \end{aligned}$$

by no feedback in the first factor and memorylessness in the second. Divide by $p(x^n)$. $\square$

# The running example: the binary symmetric channel




$\mathrm{BSC}(p)$: $\mathcal{X} = \mathcal{Y} = \{0,1\}$, each bit arrives intact with probability $1-p$ and flipped with probability $p$.

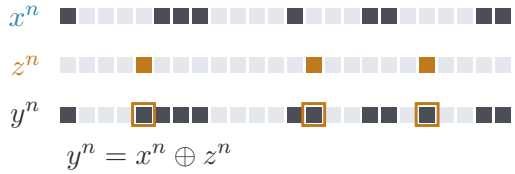
Equivalently, $n$ uses act as

$$Y^n = X^n \oplus Z^n,$$

with $Z_1, \ldots, Z_n$ i.i.d. $\mathrm{Ber}(p)$ and independent of $X^n$. Here $\oplus$ is addition modulo 2 in each coordinate, and the *noise pattern* $Z^n$ marks the flips.

Throughout: $p = 0.11$. (For $p > \frac{1}{2}$ complement the output, $p = \frac{1}{2}$ is useless.)

## One block of $24$ bits through $\mathrm{BSC}(0.11)$




- The number of flips $K$ has law $\mathrm{Bin}(24, 0.11)$, with mean $24 \cdot 0.11 = 2.64$.
- In this sample $K = 3$: the outlined bits of $y^n$ are the flipped ones.
- The block arrives intact with probability $0.89^{24} \approx 0.061$. Without coding, about $94\%$ of such blocks contain an error.

## Rate and error probability

**Definition 1.4 (Rate, error probabilities).** The *rate* of an $(M, n)$ code is $R = \frac{1}{n} \log M$, in message bits per channel use. The *conditional error* of message $w$ is $\lambda_w = \mathbb{P}(g(Y^n) \neq w \mid X^n = x^n(w))$, the *average error* is $P_e^{(n)} = \frac{1}{M} \sum_w \lambda_w$, the *maximal error* is $\lambda^{(n)} = \max_w \lambda_w \geq P_e^{(n)}$.

If $W$ is uniform on $\mathcal{W}$ and $\hat{W} = g(Y^n)$, then $P_e^{(n)} = \mathbb{P}(\hat{W} \neq W)$.

**Example 1.5 (Uncoded transmission).** $M = 2^n$, codewords all binary words, $g$ the identity: $R = 1$ and $\lambda_w = 1 - (1-p)^n \to 1$. For $n = 24$, $\lambda^{(n)} = P_e^{(n)} \approx 0.94$.

**Achievable rate:** $R$ is *achievable* if there are $(\lceil 2^{nR} \rceil, n)$ codes with $\lambda^{(n)} \to 0$ as $n \to \infty$.

## Capacity

**Definition 1.6 (Information capacity).** The *capacity* of a discrete memoryless channel is

$$C = \max_{p(x)} I(X;Y),$$

the maximum over all input distributions $p(x)$ on $\mathcal{X}$, with $I(X;Y)$ computed for the joint law $p(x)\,p(y \mid x)$ of one input and one output.

- The input law is all the sender controls in one use. $I(X;Y)$ measures how much one output reveals about one input.
- $I(X;Y)$ is a continuous, concave function of $p(x)$, so the maximum exists.
- $0 \le C \le \min\{\log|\mathcal{X}|, \log|\mathcal{Y}|\}$.

The theorem will give this single-letter quantity an operational meaning: the largest rate with vanishing error.

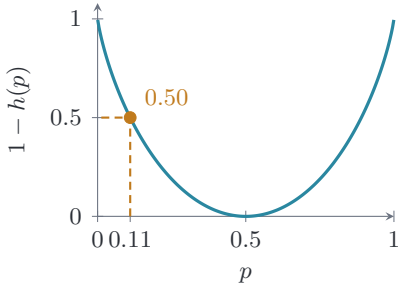
## Capacity of the binary symmetric channel

For any input law $X \sim \mathrm{Ber}(q)$:

$$\begin{aligned} I(X;Y) &= H(Y) - H(Y \mid X) \\ &= H(Y) - h(p) \\ &\le 1 - h(p). \end{aligned}$$

- $H(Y \mid X = x) = h(p)$ for both inputs $x$.
- $Y$ is binary: $H(Y) \le 1$, with equality iff $Y$ is uniform.
- The input $\mathrm{Ber}(\frac{1}{2})$ makes $Y$ uniform: $\mathbb{P}(Y = 1) = \frac{1}{2}(1-p) + \frac{1}{2}p = \frac{1}{2}$.

Hence $C = 1 - h(p)$ for $\mathrm{BSC}(p)$.




$$\begin{aligned} h(0.11) &= 0.11 \log \tfrac{1}{0.11} + 0.89 \log \tfrac{1}{0.89} \\ &\approx 0.4999, \end{aligned}$$

so $C \approx 0.50$: one message bit per two channel bits.

## Pause and think

Only $11\%$ of the bits are flipped, so $89\%$ arrive intact. Why is the capacity about $0.50$, and not $0.89$?

**Answer.** The receiver does not know *which* bits are intact. Locating the flips costs $h(p) \approx 0.50$ bits per symbol, and $C = 1 - h(p)$ subtracts exactly that.

- If the damaged positions were marked (bits erased, not flipped), the capacity would indeed be $1 - 0.11 = 0.89$.
- Shannon's own example (1948, Section 12): with $1\%$ flips the naive answer is $0.99$, the capacity is $1 - h(0.01) \approx 0.919$.

## Shannon's channel coding theorem

**Theorem 1.7 (Channel coding theorem).** Let a discrete memoryless channel with capacity $C$ be used without feedback.

(a) *Direct part.* Every rate $R < C$ is achievable: there are $(\lceil 2^{nR} \rceil, n)$ codes with $\lambda^{(n)} \to 0$ as $n \to \infty$.

(b) *Converse.* If $(\lceil 2^{nR} \rceil, n)$ codes have $P_e^{(n)} \to 0$, then $R \leq C$.

- A positive rate and a vanishing error, at the same time. The price is paid in block length $n$.
- Today: (a). Next lecture: (b). Together, $C$ is the supremum of achievable rates, for both error criteria.
- History: Theorem 11 of Shannon (1948), proved by averaging over random codes. Rigorous proof of (a) with the maximal error: Feinstein (1954).

## Summary of part 1

- A memoryless channel used without feedback acts on blocks by the product formula $p(y^n \mid x^n) = \prod_i p(y_i \mid x_i)$.
- $\mathrm{BSC}(p)$: $Y^n = X^n \oplus Z^n$, with noise $Z^n$ i.i.d. $\mathrm{Ber}(p)$, independent of the input.
- A code has a rate $R = \frac{1}{n} \log M$ and error probabilities $\lambda_w$, $P_e^{(n)}$, $\lambda^{(n)}$.
- $C = \max_{p(x)} I(X;Y)$. For $\mathrm{BSC}(0.11)$, $C = 1 - h(0.11) \approx 0.50$.
- The promise: every $R < C$ is achievable.

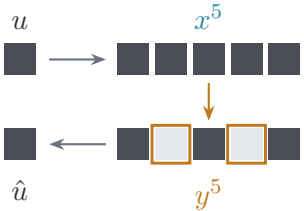
**Next:** the obvious way to fight noise, and why it cannot keep a positive rate.

Part 2

# Reliability by repetition, and its price

The obvious defence against noise, its exact error, and why it cannot keep a positive rate.

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**Example 2.1 (Repetition code).** Let $n$ be odd. The *repetition code* of length $n$ is the $(2, n)$ code with codewords $0^n$ and $1^n$ and the *majority-vote* decoder, which returns the bit value that fills more than half of the received block. Its rate is $R = \frac{1}{n}$.

Here a $1$ is sent five times and copies $2$ and $4$ are flipped. The block $y^5$ still holds three ones against two zeros, so the vote reads $\hat{u} = 1$: two flips are corrected.

## When does the vote fail?

The vote errs exactly when more than half of the bits flip. With $K \sim \mathrm{Bin}(n, p)$ flips,

$$\lambda_1 = \lambda_2 = P_e^{(n)} = \mathbb{P}\Big(K \geq \frac{n+1}{2}\Big) = \sum_{k=(n+1)/2}^{n} \binom{n}{k} p^k (1-p)^{n-k}.$$

For $p = 0.11$:

- $n = 3$: $3p^2(1-p) + p^3 \approx 0.0323 + 0.0013 \approx 0.0336$.
- $n = 5$: $\binom{5}{3}p^3(1-p)^2 + \binom{5}{4}p^4(1-p) + p^5 \approx 0.01054 + 0.00065 + 0.00002 \approx 0.0112$.

$$P_e^{(n)} \approx 0.011 \ \ll \ p = 0.11 \quad \text{(the error of one unprotected bit)}.$$

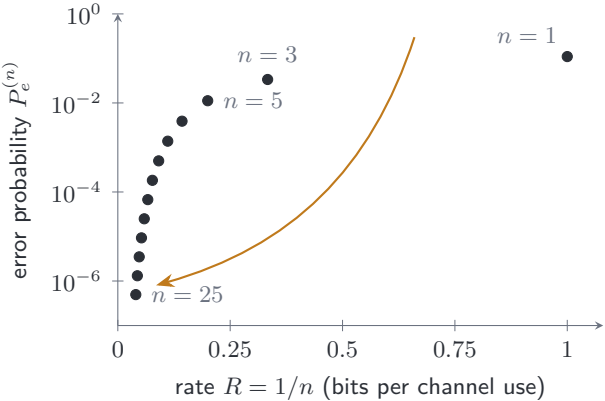
**Proposition 2.2 (Repetition codes).** Let $0 \le p \le \frac{1}{2}$ and $n$ odd. The repetition code on $\mathrm{BSC}(p)$ has $P_e^{(n)} \le \big(4p(1-p)\big)^{n/2}$.

*Proof.* For $k \ge n/2$, since $p \le 1-p$, $p^k(1-p)^{n-k} = \big(p(1-p)\big)^{n/2}\big(\frac{p}{1-p}\big)^{k-n/2} \le \big(p(1-p)\big)^{n/2}$. Insert in the sum for $P_e^{(n)}$ and use $\sum_k \binom{n}{k} = 2^n$. □

| length $n$ | 1 | 3 | 5 | 11 | 25 |
|---|---|---|---|---|---|
| rate $R = 1/n$ | 1 | 0.333 | 0.2 | 0.091 | 0.04 |
| error $P_e^{(n)}$ | 0.11 | 0.034 | 0.011 | $5.0 \times 10^{-4}$ | $4.95 \times 10^{-7}$ |
| bound $0.626^n$ | 0.626 | 0.245 | 0.096 | 0.0058 | $8.1 \times 10^{-6}$ |

For $p = 0.11$, $4p(1-p) = 0.3916$ and $\sqrt{0.3916} \approx 0.626$. The error falls geometrically, but only because the rate $1/n$ falls to $0$.

# Reliability bought with rate




Each dot is one repetition code, $n = 1, 3, \ldots, 25$.

The dots reach small errors only by sliding towards $R = 0$.

Repetition buys reliability by giving up rate.

## Pause and think

Keep the rate fixed at $\frac{1}{3}$: repeat each of $k$ message bits three times and decode each bit by majority. What happens to the probability that the *whole* message is decoded correctly as $k$ grows?

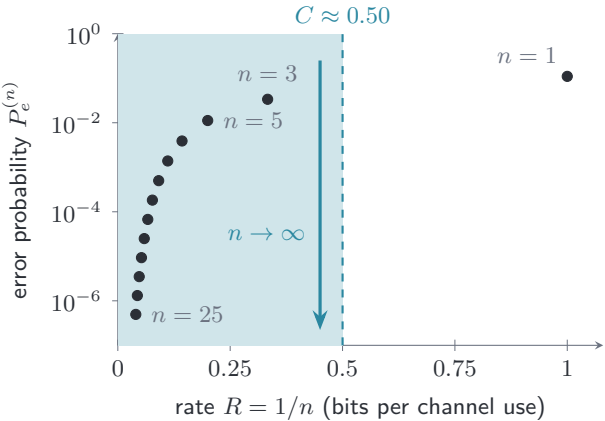
**Answer.** Each bit is wrong with probability $0.0336$, independently, so

$$\mathbb{P}(\text{all } k \text{ bits correct}) = (1 - 0.0336)^k \xrightarrow[k\to\infty]{} 0.$$

For $k = 100$ ($n = 300$): $0.9664^{100} \approx 0.033$, a block error of about $0.97$.

At a fixed rate, protecting bits one at a time makes the block error tend to $1$. Shannon's codes must code across the whole block.

# Shannon's claim is of a different kind




At any fixed rate $R < C \approx 0.50$ (shaded), the error can be made as small as we like by taking $n$ large.

The price is paid in block length and in decoding effort, not in rate.

Preview of part 6: for $n = 1000$ there is a code of rate $0.249$ with maximal error below $1.4 \times 10^{-6}$. Repetition needs rate about $0.04$ for errors of this size.

## Summary of part 2

- The repetition code of length $n$ has rate $\frac{1}{n}$ and error $\mathbb{P}(K \geq \frac{n+1}{2}) \leq (4p(1-p))^{n/2}$.
- For $p = 0.11$: error $0.034$ at $n = 3$, $0.011$ at $n = 5$, below $10^{-6}$ first at $n = 25$, where the rate is $0.04$.
- Protecting bits one at a time cannot keep a positive rate and a small block error at the same time.
- Shannon: fixed $R < C$, error $\to 0$ as $n \to \infty$.

**Next:** to code across a long block, first understand what a long noise pattern looks like.

Part 3

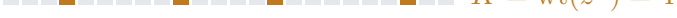
# Noise is predictable in bulk

A long noise pattern is almost surely one of a small, nearly uniform set of typical patterns.

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## The noise on its own

**Convention for parts 3 to 6**
$\mathrm{BSC}(p)$ with $0 < p < \frac{1}{2}$, so $Y^n = X^n \oplus Z^n$ with $Z^n$ i.i.d. $\mathrm{Ber}(p)$, independent of $X^n$. The *Hamming weight* $\mathrm{wt}(z^n)$ is the number of ones of $z^n$. The *Hamming distance* $d(x^n, y^n) = \mathrm{wt}(x^n \oplus y^n)$ is the number of positions where $x^n$ and $y^n$ differ.

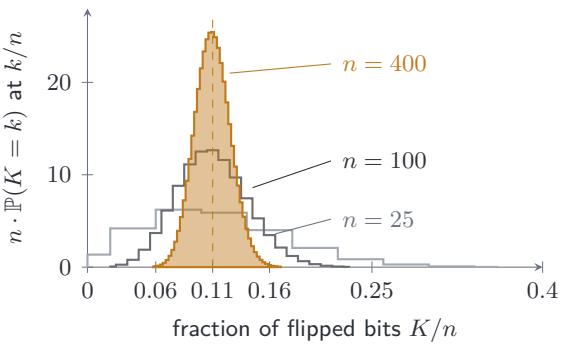



The number of flips $K = \mathrm{wt}(Z^n)$ has law $\mathrm{Bin}(n, p)$:

$$\mathbb{E}K = np, \qquad \mathrm{Var}\, K = np(1-p).$$

The noise does not depend on what is sent: we can study it on its own.

# The fraction of flips concentrates




$K/n$ has mean $p$ and standard deviation $\sqrt{p(1-p)/n}$: $0.063$ for $n = 25$, $0.031$ for $n = 100$, $0.016$ for $n = 400$.

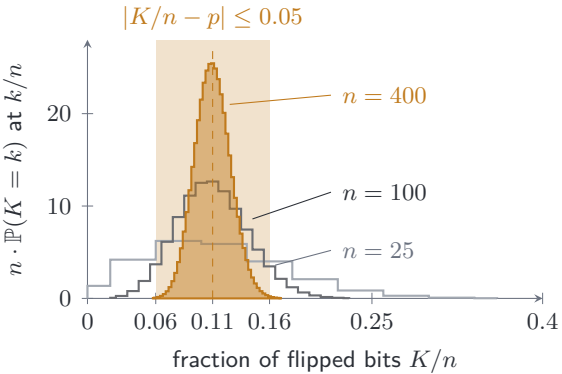
**Proposition 3.1 (Concentration).** For $K \sim \mathrm{Bin}(n, p)$ and $\varepsilon > 0$,

$$\mathbb{P}\Big(\Big|\frac{K}{n} - p\Big| \geq \varepsilon\Big) \leq \frac{p(1-p)}{n\varepsilon^2}.$$

*Proof.* Chebyshev's inequality for $K/n$, whose variance is $p(1-p)/n$. □

# Typical noise patterns

**Definition 3.2 (Typical noise patterns).** For $\varepsilon > 0$, the patterns whose fraction of flips is within $\varepsilon$ of $p$: $T_\varepsilon^{(n)} = \{z^n \in \{0,1\}^n : \ |\mathrm{wt}(z^n)/n - p| \le \varepsilon\}$.




With $\varepsilon = 0.05$ the band is $0.06 \le K/n \le 0.16$.

Chebyshev: $\frac{p(1-p)}{n\varepsilon^2} = \frac{39.16}{n}$, so $\mathbb{P}(Z^n \in T_\varepsilon^{(n)}) \ge 0.61$ for $n = 100$, and $\ge 0.902$ for $n = 400$.

Exact: $0.645$ $(n = 25)$, $0.924$ $(n = 100)$, $0.9989$ $(n = 400)$.

## How many patterns have exactly $k$ flips?

**Lemma 3.3 (One binomial coefficient).** For integers $n \geq 1$ and $0 \leq k \leq n$,
$\dfrac{2^{nh(k/n)}}{n+1} \leq \binom{n}{k} \leq 2^{nh(k/n)}$.

*Proof.* Let $q = k/n$ and $t_j = \binom{n}{j} q^j (1-q)^{n-j}$, $j = 0, \ldots, n$: the $n+1$ probabilities of $\mathrm{Bin}(n, q)$.

- Each $t_j \leq 1$, and the largest is at least $\frac{1}{n+1}$.
- The largest is $t_k$. For $k \in \{0, n\}$ the law is a point mass at $k$. For $0 < k < n$, $\frac{t_{j+1}}{t_j} = \frac{(n-j)k}{(j+1)(n-k)}$ exceeds $1$ for $j < k$ and is at most $1$ for $j \geq k$.
- $t_k = \binom{n}{k} 2^{-nh(q)}$, since $\log\left(q^k (1-q)^{n-k}\right) = n\big(q \log q + (1-q)\log(1-q)\big) = -nh(q)$. □

**Example.** $n = 100$, $k = 11$: $2^{43.3} \leq \binom{100}{11} \approx 2^{47.0} \leq 2^{50.0}$.

## The volume of a Hamming ball

**Lemma 3.4 (Volume bound).** For $0 < q \le \frac{1}{2}$ and every $n \ge 1$, $\sum_{0 \le k \le nq} \binom{n}{k} \le 2^{nh(q)}$.

*Proof.* By the binomial theorem,

$$1 = \big(q + (1-q)\big)^n \;\ge\; \sum_{0 \le k \le nq} \binom{n}{k} (1-q)^n \Big(\frac{q}{1-q}\Big)^k \;\ge\; q^{nq}(1-q)^{n(1-q)} \sum_{0 \le k \le nq} \binom{n}{k}$$

because $q \le 1-q$ makes $(q/(1-q))^k$ decrease in $k$, so it is at least $(q/(1-q))^{nq}$ for $k \le nq$. Finally $q^{nq}(1-q)^{n(1-q)} = 2^{-nh(q)}$. □

The sum counts the words within distance $nq$ of a fixed word: the *Hamming ball* of radius $nq$. It is used twice: to count typical noise (now), and to bound the chance that a wrong codeword lies close to the received word (part 6).

## How many typical patterns?

**Proposition 3.5 (Size of the typical set).** Let $0 < p < \frac{1}{2}$.

(a) If $k_n$ is an integer nearest to $np$, then $\frac{1}{n}\log\binom{n}{k_n} \to h(p)$.

(b) If $\varepsilon > 0$ and $p + \varepsilon \le \frac{1}{2}$, then $|T_\varepsilon^{(n)}| \le 2^{nh(p+\varepsilon)}$ for every $n$, and $\frac{1}{n}\log|T_\varepsilon^{(n)}| \to h(p+\varepsilon)$.

*Proof.* (a) By the lemma, $h(\frac{k_n}{n}) - \frac{\log(n+1)}{n} \le \frac{1}{n}\log\binom{n}{k_n} \le h(\frac{k_n}{n})$, and $|\frac{k_n}{n} - p| \le \frac{1}{2n}$, $h$ continuous.

(b) Every pattern in $T_\varepsilon^{(n)}$ has at most $n(p+\varepsilon)$ ones: the volume bound with $q = p + \varepsilon$. For the limit, let $m_n = \lfloor n(p+\varepsilon) \rfloor$. For $n \ge \frac{1}{2\varepsilon}$ all $\binom{n}{m_n}$ patterns of weight $m_n$ are typical, and the lemma gives $\frac{1}{n}\log|T_\varepsilon^{(n)}| \ge h(\frac{m_n}{n}) - \frac{\log(n+1)}{n} \to h(p+\varepsilon)$. □

## Counting for $p = 0.11$

| $n$ | 25 | 100 | 400 | 1600 |
|---|---|---|---|---|
| $k_n$ | 3 | 11 | 44 | 176 |
| $\frac{1}{n}\log\binom{n}{k_n}$ | 0.447 | 0.470 | 0.490 | 0.497 |

$\longrightarrow\ h(0.11) \approx 0.4999$

- Patterns with exactly $np$ flips: about $2^{nh(p)}$. The whole band: about $2^{nh(p+\varepsilon)}$, larger. The two exponents meet as $\varepsilon \to 0$.
- $n = 100$, $\varepsilon = 0.05$: the band holds the patterns with $6$ to $16$ flips, $|T_{0.05}^{(100)}| = \sum_{k=6}^{16}\binom{100}{k} \approx 2^{60.5}$ (bound $2^{63.4}$), and carries probability $0.924$.
- Yet it is a fraction $2^{60.5}/2^{100} = 2^{-39.5} \approx 1.3 \times 10^{-12}$ of all patterns. In general $|T_{\varepsilon}^{(n)}|/2^n \le 2^{-n(1-h(p+\varepsilon))} \to 0$, and $2^{nh(p)}/2^n = 2^{-n(1-h(p))} \to 0$.

## Pause and think

Which single noise pattern $z^{100}$ is the most likely? Is it typical?

**Answer.** The all-zero pattern (no flip), with probability $0.89^{100} \approx 8.7 \times 10^{-6}$. It is *not* typical: it has $0$ flips, not about $11$.

A pattern with $11$ flips has probability $0.11^{11}\, 0.89^{89} = 2^{-100\, h(0.11)} \approx 8.9 \times 10^{-16}$, about $10^{10}$ times smaller. But there are $\binom{100}{11} \approx 2^{47}$ of them:

$$\mathbb{P}(K = 11) = \binom{100}{11} 0.11^{11}\, 0.89^{89} \approx 0.127.$$

Typical patterns win by their number, not by their individual probability.

## The probability of one typical pattern

A pattern $z^n$ with $K = \mathrm{wt}(z^n)$ flips has probability $p(z^n) = p^K(1-p)^{n-K}$. With $a = \log\frac{1-p}{p}$:

$$-\tfrac{1}{n}\log p(z^n) = -\tfrac{K}{n}\log p - (1-\tfrac{K}{n})\log(1-p) = -\log(1-p) + a\,\tfrac{K}{n},$$
$$h(p) = -\log(1-p) + a\,p.$$

Subtracting,

$$p(z^n) = 2^{-n\left(h(p)+\left(\frac{K}{n}-p\right)a\right)}.$$

So for every $z^n \in T_\varepsilon^{(n)}$: $\quad 2^{-n(h(p)+\varepsilon a)} \le p(z^n) \le 2^{-n(h(p)-\varepsilon a)}$. $\quad$ For $p = 0.11$, $a \approx 3.02$.

All typical patterns are about equally likely, each about $2^{-nh(p)}$: the probability is split almost evenly among them.

## The asymptotic equipartition property

**Theorem 3.6 (AEP).** Let $X_1, X_2, \dots$ be i.i.d. on a finite set with law $p(x)$ and entropy $H$. Then $-\frac{1}{n}\log p(X^n) \to H$ in probability. The *typical set* $A_\varepsilon^{(n)}(X) = \{x^n : 2^{-n(H+\varepsilon)} \le p(x^n) \le 2^{-n(H-\varepsilon)}\}$ satisfies (a) $\mathbb{P}(X^n \in A_\varepsilon^{(n)}(X)) \to 1$, (b) $|A_\varepsilon^{(n)}(X)| \le 2^{n(H+\varepsilon)}$ for all $n$, (c) $|A_\varepsilon^{(n)}(X)| \ge (1-\varepsilon)2^{n(H-\varepsilon)}$ for all large $n$.

*Proof.* $-\frac{1}{n}\log p(X^n) = \frac{1}{n}\sum_i L_i$ with $L_i = -\log p(X_i)$ i.i.d. of mean $H$: the weak law gives (a). (b) $1 \ge \sum_{A} p(x^n) \ge |A|\, 2^{-n(H+\varepsilon)}$. (c) $1 - \varepsilon \le \mathbb{P}(A) \le |A|\, 2^{-n(H-\varepsilon)}$ for large $n$. □

For the BSC noise, $H = h(p)$ and the previous frame gives $A_\varepsilon^{(n)}(Z) = T_{\varepsilon/a}^{(n)}$: typical in the sense of the AEP means having close to $np$ flips.

## Summary of part 3

- The fraction of flips concentrates: $\mathbb{P}(|K/n - p| \geq \varepsilon) \leq p(1-p)/(n\varepsilon^2)$.
- A Hamming ball of radius $qn$, $q \leq \frac{1}{2}$, has at most $2^{nh(q)}$ words.
- Typical patterns: about $2^{nh(p)}$ with exactly $np$ flips, at most $2^{nh(p+\varepsilon)}$ in the band. A vanishing fraction of $\{0,1\}^n$, carrying almost all the probability.
- Each typical pattern has probability about $2^{-nh(p)}$ (AEP).

**Next:** each codeword drags a cloud of typical outputs with it. How many clouds fit?

Part 4

# How many messages can the noise keep apart?

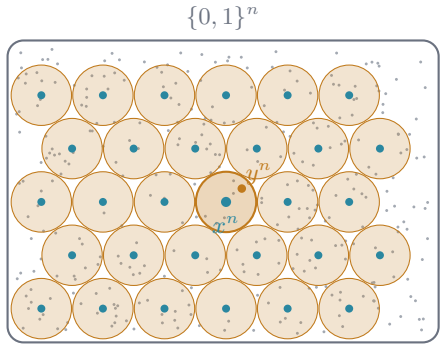
A counting argument shows which rates can possibly work, and what the direct part must achieve.

1 Reliable transmission of information
2 Reliability by repetition, and its price
3 Noise is predictable in bulk
**4 How many messages can the noise keep apart?**
5 A random codebook and a threshold decoder
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# How many noise clouds fit?

- The received word lives in $\{0,1\}^n$, a set of $2^n$ words.
- Sending $x^n$ gives $y^n = x^n \oplus z^n$, and $z^n \in T_\varepsilon^{(n)}$ with high probability. So $y^n$ lies in the *noise cloud* $x^n \oplus T_\varepsilon^{(n)}$, of $|T_\varepsilon^{(n)}| \approx 2^{nh(p)}$ words ($\varepsilon$ small).
- Distinguishable messages need clouds that barely overlap.
- Disjoint clouds share the $2^n$ words: $M \cdot 2^{nh(p)} \lesssim 2^n$, so

$$R = \tfrac{1}{n} \log M \lesssim 1 - h(p) = C.$$



## What the count says, and what it does not

- $n = 1000$: each cloud holds about $2^{500}$ words, so at most about $2^{1000}/2^{500} = 2^{500}$ disjoint clouds fit: rate $0.50$.
- With the honest size of the band, $|T_{0.05}^{(1000)}| \approx 2^{1000\,h(0.16)} = 2^{634}$, the count gives $2^{366}$ clouds: rate $1 - h(0.16) \approx 0.37$. This number returns in part 6.
- The cloud is a thin shell: its words lie at distance between $n(p-\varepsilon)$ and $n(p+\varepsilon)$ from $x^n$, and most of them near the outer radius.

**Only a heuristic**
Reliable decoding needs clouds whose overlaps carry little probability, not disjoint clouds, and the error may be positive. The strong converse (next lecture) turns this count into a theorem by counting decoding regions.

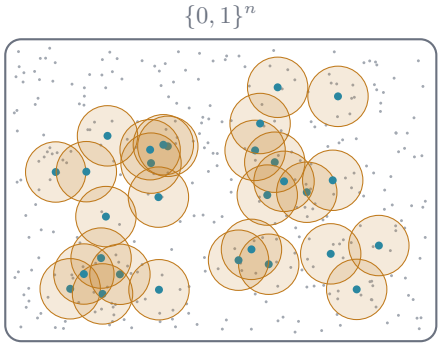
**The task of the direct part:** for every $R < C$, place $2^{nR}$ codewords whose clouds barely overlap.

## Pause and think

Design a code with $n = 1000$ and $M = 2^{250}$ codewords ($R = 0.25$) such that the received word rarely falls near a wrong codeword. How would you place the codewords?

**Shannon's answer:** do not design the code at all. Choose the codewords at random.

A uniform random word lies within distance $n(p + \varepsilon)$ of the received word with probability at most $2^{-n(1-h(p+\varepsilon))} = 2^{-365.7}$. With $2^{250}$ such words, the expected number of intruders is at most $2^{250-365.7} = 2^{-115.7}$.




Random clouds do overlap. What matters is that the overlaps carry little probability.

## Summary of part 4

- Each codeword carries a noise cloud of about $2^{nh(p)}$ words (at most $2^{nh(p+\varepsilon)}$ with slack $\varepsilon$) inside a space of $2^n$ words.
- Disjoint clouds: $M \lesssim 2^{n(1-h(p))}$, so rates above $C$ look impossible. This is a heuristic for the converse, not a proof.
- The direct part must place $2^{nR}$ codewords, $R < C$, whose clouds barely overlap. Shannon chooses them at random.

**Next:** the random codebook and the rule that decodes it.

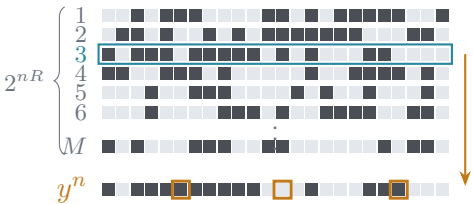
Part 5

# A random codebook and a threshold decoder

Shannon's construction: draw the codebook at random, decode by a distance threshold.

1 Reliable transmission of information
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# A random codebook

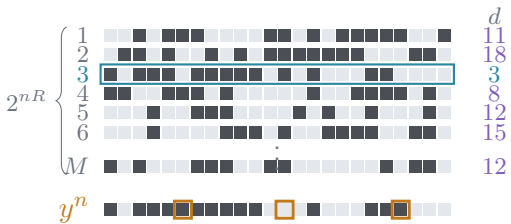



**Definition 5.1 (Random codebook).** Fix $n$, a rate $R < C$ and $M = \lceil 2^{nR} \rceil$. The codebook $\mathcal{C} = (X^n(1), \ldots, X^n(M))$ has all its $Mn$ bits i.i.d. $\mathrm{Ber}(\frac{1}{2})$. It is drawn once and given to both sender and receiver.

To send $w$, transmit its row $X^n(w)$. The channel flips about $np$ of its bits (here $n = 24$, three flips) and the receiver gets $y^n$.

For $n = 1000$ and $R = 0.25$ the codebook has $2^{250}$ rows: a proof device, not a practical code.

# The threshold decoder




The rule is a one-sided form of joint typicality decoding (part 6).

**Definition 5.2 (Threshold decoder).** Given $y^n$, output $\hat{w}$ if $x^n(\hat{w})$ is the *only* codeword with $d(x^n(\hat{w}), y^n) \leq n(p+\varepsilon)$. Otherwise output $1$, counted as an error.

Here $n(p+\varepsilon) = 24 \cdot 0.16 = 3.84$.

Row $3$ is at distance $3$: it differs from $y^n$ only where the noise flipped bits. The other rows are at distances $8$ to $18$, near $n/2 = 12$.

Only row $3$ passes: $\hat{w} = 3$.

## Distances to the received word

**Lemma 5.3 (Two distance laws).** Let the codebook be random, let message $w$ be sent over $\mathrm{BSC}(p)$, and let $Y^n$ be the output.

(a) $d(X^n(w), Y^n) = \mathrm{wt}(Z^n)$, which has law $\mathrm{Bin}(n, p)$.

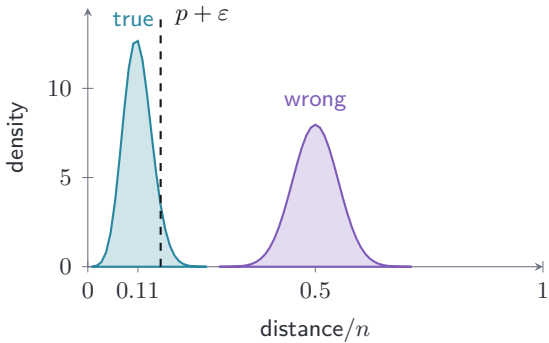
(b) For $w' \neq w$, the codeword $X^n(w')$ is uniform on $\{0,1\}^n$ and independent of $Y^n$, and $d(X^n(w'), Y^n)$ has law $\mathrm{Bin}(n, \frac{1}{2})$.

*Proof.* (a) $X^n(w) \oplus Y^n = Z^n$.

(b) $Y^n = X^n(w) \oplus Z^n$ is a function of $(X^n(w), Z^n)$, which is independent of $X^n(w')$. For a fixed $y^n$, the map $x^n \mapsto x^n \oplus y^n$ is a bijection of $\{0,1\}^n$, so $X^n(w') \oplus y^n$ is uniform, and its weight $d(X^n(w'), y^n)$ is $\mathrm{Bin}(n, \frac{1}{2})$, whatever $y^n$ is. □

The true codeword sits at distance about $np$, a wrong one at about $n/2$.

# Two laws, one threshold



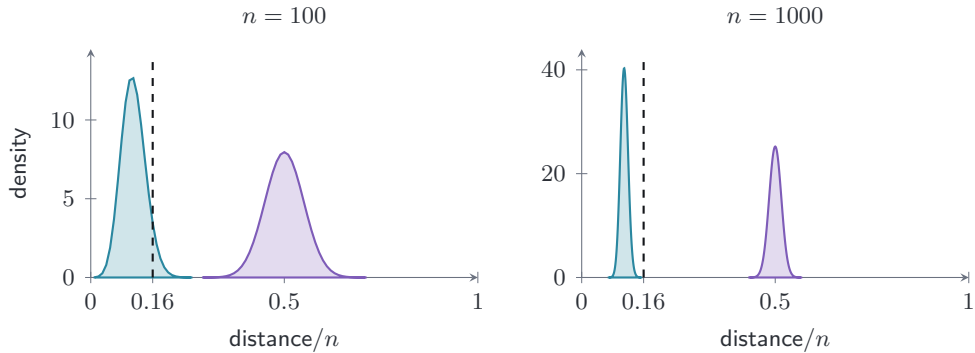

$n = 100$, distances divided by $n$.

True codeword: mean $p = 0.11$, standard deviation $\sqrt{p(1-p)/n} \approx 0.031$.

Wrong codeword: mean $0.5$, standard deviation $\sqrt{1/(4n)} = 0.05$.

The threshold $p + \varepsilon = 0.16$ sits in the gap: $1.6$ standard deviations above the true mean, $6.8$ below the wrong one.

## The gap becomes a wall




Both spreads shrink like $1/\sqrt{n}$. At $n = 1000$ the standard deviations are $0.0099$ and $0.0158$: the threshold $0.16$ is $5.1$ standard deviations above the true mean and $21.5$ below the wrong one.

## Pause and think

Why not put the threshold in the middle of the gap, at $0.3$, where both laws are far away?

**Answer.** A threshold $qn$ accepts a wrong codeword when it falls in a Hamming ball of radius $qn$ around $y^n$. By the volume bound this has probability at most $2^{-n(1-h(q))}$, and with $2^{nR}$ wrong codewords the union bound needs $R < 1 - h(q)$.

| | | | |
|---|---|---|---|
| threshold $q$ | $0.3$ | $0.16$ | $q \to p = 0.11$ |
| largest rate $1 - h(q)$ | $0.12$ | $0.37$ | $C \approx 0.50$ |

To reach rates near capacity the threshold must hug $p$. The price is a true codeword that falls outside more often, paid for with a larger $n$.

## Summary of part 5

- Random codebook: $M = \lceil 2^{nR} \rceil$ rows of i.i.d. uniform bits, known to both ends.
- Threshold decoder: the unique codeword within distance $n(p + \varepsilon)$ of $y^n$.
- The true codeword is at distance $\mathrm{Bin}(n, p)$, about $np$. A wrong codeword is uniform and independent of $y^n$, at distance $\mathrm{Bin}(n, \frac{1}{2})$, about $n/2$.
- The threshold $0.16$ sits between $0.11$ and $0.5$, and the gap widens like $\sqrt{n}$ in units of standard deviation.

**Next:** turn the picture into a bound on the error probability.

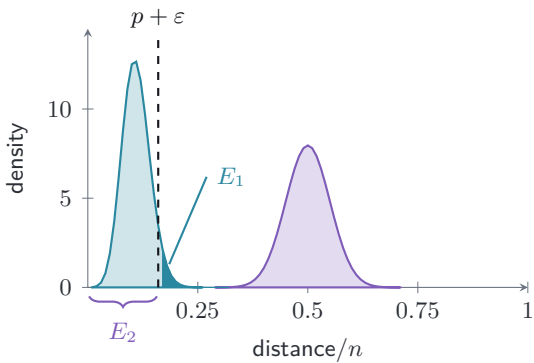
Part 6

# Why the error vanishes

Two error events, the law of large numbers, a union bound, and two steps from a good ensemble to a good code.

1 Reliable transmission of information
2 Reliability by repetition, and its price
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# Two ways to fail




Message $w$ is sent. The decoder can fail in only two ways:

- $E_1 = \{d(X^n(w), Y^n) > n(p+\varepsilon)\}$: the true codeword is rejected.
- $E_2 = \{d(X^n(w'), Y^n) \le n(p+\varepsilon)$ for some $w' \neq w\}$: some wrong codeword is accepted.

If neither occurs, $x^n(w)$ is the only codeword within the threshold and the decoder outputs $w$. Averaging over the codebook too,

$$\mathbb{E}_{\mathcal{C}}\left[\lambda_w(\mathcal{C})\right] \le \mathbb{P}(E_1) + \mathbb{P}(E_2).$$

## $E_1$: the law of large numbers

By the lemma of part 5, $d(X^n(w), Y^n) = K \sim \mathrm{Bin}(n, p)$, so

$$\mathbb{P}(E_1) = \mathbb{P}\big(K > n(p+\varepsilon)\big) \le \mathbb{P}\Big(\Big|\frac{K}{n} - p\Big| \ge \varepsilon\Big) \le \frac{p(1-p)}{n\varepsilon^2} \xrightarrow[n\to\infty]{} 0.$$

| | event | Chebyshev bound | exact $\mathbb{P}(E_1)$ |
|---|---|---|---|
| $n = 100$ | $K \ge 17$ | $0.39$ | $0.046$ |
| $n = 1000$ | $K > 160$ | $0.039$ | $6.9 \times 10^{-7}$ |

About $5\%$ at $n = 100$, below one in a million at $n = 1000$.

$\mathbb{P}(E_1)$ does not depend on the number of codewords: it is the same at every rate.

## $E_2$, one wrong codeword at a time

Fix $y^n$. A wrong codeword $\tilde{X}^n$ is uniform on $\{0,1\}^n$ (part 5), so

$$\mathbb{P}\big(d(\tilde{X}^n, y^n) \le n(p+\varepsilon)\big) = \frac{\#\{x^n : d(x^n, y^n) \le n(p+\varepsilon)\}}{2^n} = 2^{-n} \sum_{0 \le k \le n(p+\varepsilon)} \binom{n}{k}$$
$$\le 2^{-n}\, 2^{nh(p+\varepsilon)} = 2^{-n(1-h(p+\varepsilon))},$$

by the volume bound with $q = p + \varepsilon \le \frac{1}{2}$. The bound holds for every $y^n$, hence also for the random $Y^n$, which is independent of $\tilde{X}^n$.

**Example.** $n = 100$, $\varepsilon = 0.05$: the exact value is $2^{-100} \sum_{k \le 16} \binom{100}{k} \approx 1.3 \times 10^{-12}$, the bound is $2^{-36.57} \approx 9.8 \times 10^{-12}$. The volume bound loses a factor below $8$.

## All wrong codewords: the union bound

There are $M-1$ wrong codewords. By the union bound and the previous frame,

$$\mathbb{P}(E_2) \le \sum_{w' \neq w} \mathbb{P}\big(d(X^n(w'), Y^n) \le n(p+\varepsilon)\big) \le (M-1)\, 2^{-n(1-h(p+\varepsilon))} < 2^{-n(1-h(p+\varepsilon)-R)}$$

using $M - 1 < 2^{nR}$ for $M = \lceil 2^{nR} \rceil$.

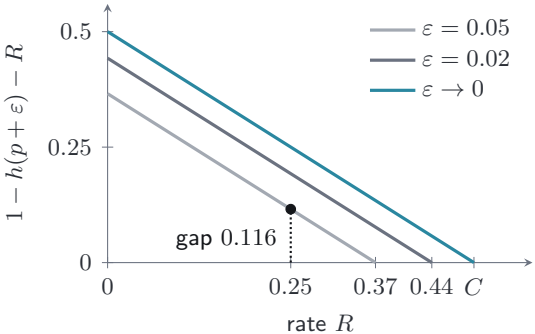
**Proposition 6.1 (Error of the random code).** Let $0 < p < \frac{1}{2}$, $\varepsilon > 0$ with $p + \varepsilon \le \frac{1}{2}$, and $M \ge 2$. For the random codebook and the threshold decoder on $\mathrm{BSC}(p)$,

$$\mathbb{E}_{\mathcal{C}}[P_e^{(n)}(\mathcal{C})] \le \mathbb{P}\big(K > n(p+\varepsilon)\big) + (M-1)\, 2^{-n(1-h(p+\varepsilon))}, \qquad K \sim \mathrm{Bin}(n, p).$$

*Proof.* The bound on $\mathbb{E}_{\mathcal{C}}[\lambda_w(\mathcal{C})]$ is the same for every $w$, so it bounds the average $\frac{1}{M}\sum_w \mathbb{E}_{\mathcal{C}}[\lambda_w(\mathcal{C})] = \mathbb{E}_{\mathcal{C}}[P_e^{(n)}(\mathcal{C})]$. □

Two terms, two origins: the law of large numbers, and counting.

# The rate condition




The bound on $\mathbb{P}(E_2)$ tends to $0$ when the gap is positive:

$$R < 1 - h(p + \varepsilon).$$

- $\varepsilon = 0.05$: $R < 0.37$.
- $\varepsilon = 0.02$: $R < 0.44$.
- $h$ is continuous: every $R < 1 - h(p) = C$ satisfies the condition for $\varepsilon$ small enough.

The slack $\varepsilon$ is the price of a fixed threshold. It vanishes in the limit, at the cost of a slower bound on $\mathbb{P}(E_1)$.

## The bound at two block lengths

$p = 0.11$, $\varepsilon = 0.05$, $R = 0.25$, threshold $0.16\,n$, $1 - h(0.16) \approx 0.366$.

| | $n = 100$, $M = 2^{25}$ | $n = 1000$, $M = 2^{250}$ |
|---|---|---|
| $\mathbb{P}(E_1)$ | $0.046$ | $6.9 \times 10^{-7}$ |
| bound on $\mathbb{P}(E_2)$ | $2^{25-36.57} = 2^{-11.57} \approx 3.3 \times 10^{-4}$ | $2^{250-365.69} \approx 1.5 \times 10^{-35}$ |
| $\mathbb{E}_{\mathcal{C}}[P_e^{(n)}(\mathcal{C})]$ | about $0.046$ | below $7 \times 10^{-7}$ |

- The exponent of the $E_2$ bound is $n$ times the gap $0.116$ of the previous frame.
- Already at $n = 100$ the wrong codewords are harmless: the error is dominated by $E_1$, the noise of the true codeword.

# From the average to one good code

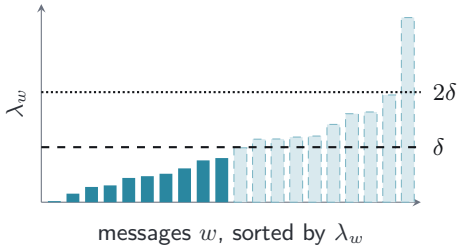



**Lemma 6.2 (Averaging).** If a random codebook satisfies $\mathbb{E}_{\mathcal{C}}[P_e^{(n)}(\mathcal{C})] \le \delta$, then some fixed codebook $\mathcal{C}^*$ has $P_e^{(n)}(\mathcal{C}^*) \le \delta$.

*Proof.* The codebook takes finitely many values. If all of them had $P_e^{(n)} > \delta$, so would their average. □

More is true. By Markov's inequality, $\mathbb{P}\big(P_e^{(n)}(\mathcal{C}) \ge t\delta\big) \le \frac{1}{t}$: at least $90\%$ of the codebooks have $P_e^{(n)} < 10\,\delta$. A codebook drawn at random is good with high probability.

# From a good average to a good worst case




**Lemma 6.3 (Expurgation).** Let an $(M, n)$ code, $M$ even, have $P_e^{(n)} \le \delta$. Keep the $M/2$ messages with the smallest $\lambda_w$, and let the decoder output a fixed kept message when it would output a discarded one. This gives an $(M/2, n)$ code with $\lambda^{(n)} \le 2\delta$ and rate $R - \frac{1}{n}$.

*Proof.* Let $\delta > 0$ (if $\delta = 0$, every $\lambda_w = 0$) and $W$ uniform, so $\mathbb{E}[\lambda_W] = P_e^{(n)} \le \delta$. Markov: $\mathbb{P}(\lambda_W \ge 2\delta) \le \frac{\delta}{2\delta} = \frac{1}{2}$, so at most $M/2$ messages have $\lambda_w \ge 2\delta$, and the $M/2$ smallest $\lambda_w$ are all below $2\delta$. A kept message is decoded correctly whenever it was before. Finally $\frac{1}{n} \log \frac{M}{2} = R - \frac{1}{n}$. □

## The direct part for the BSC

**Theorem 6.4 (Direct part for the BSC).** Let $0 < p < \frac{1}{2}$. Every rate $R < 1 - h(p)$ is achievable on $\mathrm{BSC}(p)$.

*Proof.* Choose $\varepsilon > 0$ with $p + \varepsilon \le \frac{1}{2}$ and $R < 1 - h(p+\varepsilon)$. Let $M_n = 2\lceil 2^{nR} \rceil$, so $M_n - 1 \le 2^{nR+2}$. The proposition and Chebyshev give

$$\mathbb{E}_{\mathcal{C}}\big[P_e^{(n)}(\mathcal{C})\big] \le \frac{p(1-p)}{n\varepsilon^2} + 4 \cdot 2^{-n(1-h(p+\varepsilon)-R)} =: \delta_n \xrightarrow[n\to\infty]{} 0.$$

Averaging gives an $(M_n, n)$ code with $P_e^{(n)} \le \delta_n$, expurgation a $(\lceil 2^{nR} \rceil, n)$ code with $\lambda^{(n)} \le 2\delta_n$. □

**Example 6.5 (A concrete guarantee).** $n = 1000$, $\varepsilon = 0.05$, $M = 2^{250}$: some code has $P_e^{(n)} \le \mathbb{E}_{\mathcal{C}}[P_e^{(n)}] < 7 \times 10^{-7}$. Expurgation leaves $2^{249}$ codewords: rate $0.249$, maximal error below $1.4 \times 10^{-6}$. (Repetition: rate about $0.04$.)

## Beyond the BSC: jointly typical pairs

Let $p(x)$ be an input law and $p(x,y) = p(x)p(y \mid x)$, extended to blocks as products.

**Definition 6.6 (Jointly typical set).** $A_\varepsilon^{(n)}$ is the set of pairs $(x^n, y^n)$ with $\left|-\frac{1}{n}\log p(x^n) - H(X)\right| \le \varepsilon$, $\left|-\frac{1}{n}\log p(y^n) - H(Y)\right| \le \varepsilon$ and $\left|-\frac{1}{n}\log p(x^n, y^n) - H(X,Y)\right| \le \varepsilon$.

**Theorem 6.7 (Joint AEP).** Let $(X_i, Y_i)$ be i.i.d. pairs with law $p(x,y)$.

(a) $\mathbb{P}\big((X^n, Y^n) \in A_\varepsilon^{(n)}\big) \to 1$.

(b) $|A_\varepsilon^{(n)}| \le 2^{n(H(X,Y)+\varepsilon)}$ for every $n$.

(c) If $\tilde{X}^n$, $\tilde{Y}^n$ are independent with laws $p(x^n)$ and $p(y^n)$, then $\mathbb{P}\big((\tilde{X}^n, \tilde{Y}^n) \in A_\varepsilon^{(n)}\big) \le 2^{-n(I(X;Y)-3\varepsilon)}$.

*Proof of (a), (b).* The AEP for each of the three i.i.d. sequences $(X_i)$, $(Y_i)$, $(X_i, Y_i)$, and the union bound. (b) as in the AEP: $1 \ge \sum_{A_\varepsilon^{(n)}} p(x^n, y^n) \ge |A_\varepsilon^{(n)}|\, 2^{-n(H(X,Y)+\varepsilon)}$.

## The key bound: an unrelated pair rarely looks typical

*Proof of (c).* Every pair in $A_\varepsilon^{(n)}$ has $p(x^n) \le 2^{-n(H(X)-\varepsilon)}$ and $p(y^n) \le 2^{-n(H(Y)-\varepsilon)}$. With (b),

$$\begin{aligned}\mathbb{P}\big((\tilde{X}^n, \tilde{Y}^n) \in A_\varepsilon^{(n)}\big) &= \sum_{(x^n,y^n)\in A_\varepsilon^{(n)}} p(x^n)\, p(y^n) \\ &\le 2^{n(H(X,Y)+\varepsilon)}\, 2^{-n(H(X)-\varepsilon)}\, 2^{-n(H(Y)-\varepsilon)} \\ &= 2^{-n(I(X;Y)-3\varepsilon)},\end{aligned}$$

since $H(X) + H(Y) - H(X,Y) = I(X;Y)$. □

A pair produced by the channel looks typical. A pair whose halves are unrelated, each with the right law, looks typical only with probability about $2^{-nI(X;Y)}$: mutual information measures how hard it is for an unrelated codeword to impersonate the transmitted one.

## The direct part for every memoryless channel

**Theorem 6.8 (Direct part).** Let a discrete memoryless channel with capacity $C$ be used without feedback. Every rate $R < C$ is achievable.

*Proof.* Take $p(x)$ with $I(X;Y) = C$, and $\varepsilon > 0$ with $R < C - 3\varepsilon$. Draw $M_n = 2\lceil 2^{nR} \rceil$ codewords with i.i.d. letters of law $p(x)$. Decode to the unique $\hat{w}$ with $(x^n(\hat{w}), y^n) \in A_\varepsilon^{(n)}$, and to $1$ if there is none or several. Send $w$. By the product formula, $(X^n(w), Y^n)$ is a sequence of i.i.d. pairs of law $p(x,y)$, so the true pair fails to be typical with probability $\to 0$ by (a), and $Y^n$ has law $p(y^n)$. Each $X^n(w')$, $w' \neq w$, has law $p(x^n)$ and is independent of $Y^n$, so by (c) and the union bound some wrong pair is typical with probability at most $(M_n - 1)\, 2^{-n(C-3\varepsilon)} \leq 4 \cdot 2^{-n(C-3\varepsilon-R)} \to 0$. Averaging and expurgation finish as for the BSC. □

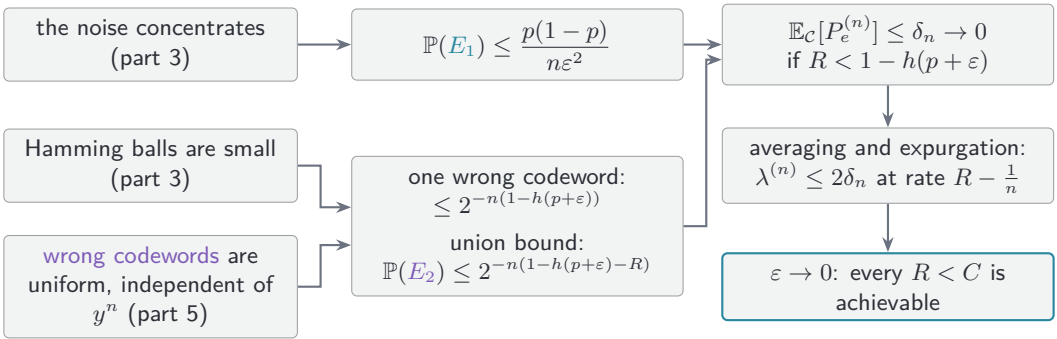
**Remark 6.9 (One-sided typicality).** On $\mathrm{BSC}(p)$ with uniform input the first two conditions always hold, and the third reads $|d(x^n, y^n)/n - p| \leq \varepsilon/a$, with $a = \log \frac{1-p}{p}$. The threshold decoder keeps only the upper half, $d \leq n(p + \varepsilon)$.

## Summary of part 6

- $\mathbb{E}_{\mathcal{C}}[P_e^{(n)}] \le \mathbb{P}(K > n(p+\varepsilon)) + (M-1)\,2^{-n(1-h(p+\varepsilon))}$: the law of large numbers for the true codeword, counting for the wrong ones.
- The bound vanishes when $R < 1 - h(p+\varepsilon)$. Letting $\varepsilon \to 0$ reaches every $R < C$.
- Averaging: one fixed good code. Expurgation: maximal error at most $2\delta$, at rate $R - \frac{1}{n}$.
- Every memoryless channel: the same proof with joint typicality, and the key bound $2^{-n(I(X;Y)-3\varepsilon)}$.

**The direct part is proved:** every rate below capacity is achievable.

# The whole argument on one slide




For a general memoryless channel: codewords with i.i.d. letters of a capacity-achieving law, joint typicality decoding, and the key bound $2^{-n(I(X;Y)-3\varepsilon)}$ in place of $2^{-n(1-h(p+\varepsilon))}$.

## What the proof gives, and what it does not

- **It gives** the existence of good codes at every rate below capacity, and even shows that a randomly drawn code is good with high probability.
- **It does not give** a practical code. Encoding needs a table of $2^{nR}$ rows, and threshold decoding a search through all of them.
- **The error decays exponentially in** $n$ at every $R < C$, not only like $1/n$ as our Chebyshev bound shows. Elias (1955) proved it for the BSC, where parity-check (linear) codes do as well as arbitrary ones. Gallager (1965) gave a short derivation for every memoryless channel.
- **History.** Shannon (1948, Theorem 11) proved the direct part by averaging over random codes. Feinstein (1954) gave a rigorous proof with the maximal error, by a different construction.

## Next lecture: the converse

Can a rate $R > C$ have vanishing error? No. Two inequalities do the work:

- **Fano:** a small error leaves little uncertainty, $H(W \mid \hat{W}) \le 1 + P_e^{(n)} \log M$.
- **Data processing:** $W \to X^n \to Y^n \to \hat{W}$, so $I(W;\hat{W}) \le I(X^n;Y^n) \le nC$ (memoryless, no feedback).

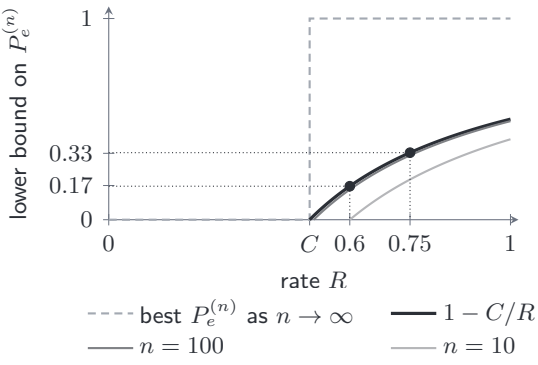
With $W$ uniform on $M = 2^{nR}$ messages:

$$nR = H(W) = H(W \mid \hat{W}) + I(W;\hat{W}) \le 1 + P_e^{(n)}\, nR + nC,$$

hence the *weak converse*

$$P_e^{(n)} \ge 1 - \frac{C}{R} - \frac{1}{nR}.$$

# Next lecture: the converse on $\mathrm{BSC}(0.11)$




$R = 0.6$: $P_e^{(n)} \geq 0.150$ ($n = 100$), $0.165$ ($n = 1000$), limit $0.167$.

$R = 0.75$: $P_e^{(n)} \geq 0.320$, $0.332$, limit $0.333$.

For the BSC the *strong converse* holds: above $C$ the error tends to $1$. Its proof counts decoding regions, and turns the packing count of part 4 into a theorem.

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